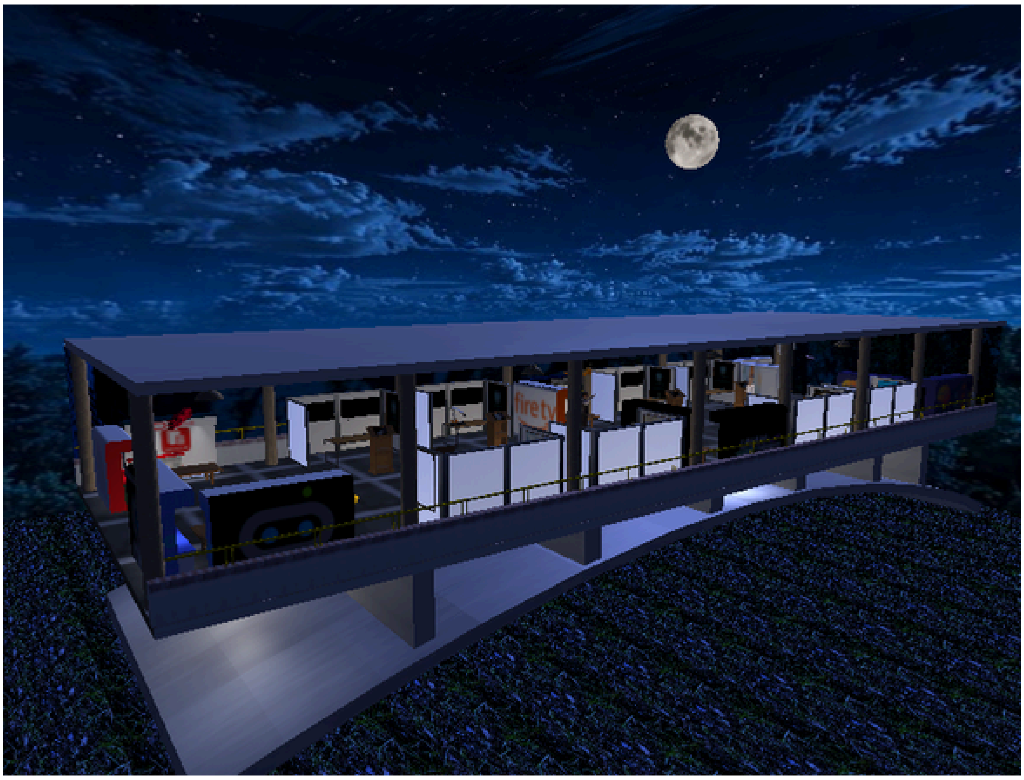
User Manual

Bridge (Buildings A-B) of the School of Engineering, UNAM



Final Project
Created by:

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Group: 05

Semester: 2026-2

Due date:
May 26, 2026

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Introduction

This document serves as the user manual for the Final Project developed for the course Computer Graphics and Human-Computer Interaction (semester 2026-2). Its purpose is to provide users with a clear and precise guide on the steps required to obtain, install, run, and operate the system, as well as to understand the functionality of each of its interactive components.

The project consists of a three-dimensional virtual environment that reproduces, for academic purposes, the bridge at the Faculty of Engineering of the National Autonomous University of Mexico. This space is commonly used during university events to host booths for companies and student associations. The application allows users to freely explore the bridge, switch between two lighting schemes (day and night), and interact with a set of six independent animations, which represent common situations during an academic fair.

This manual is intended for end users who wish to run the project on a local computer. It is recommended that you review it in its entirety before the first run, in order to avoid common configuration errors and take full advantage of all the implemented features.

System requirements

The following are the recommended hardware and software requirements for the successful implementation of the project.

Component	Recommended
Operating system	Windows 10 / 11
Processor	Intel Core i5 (9th gen.) or higher
RAM	8 GB or more
Graphics card	NVIDIA GTX 1060 / RTX 2060 or higher
Disk space	10 GB
OpenGL	Version 4.0 or higher
IDE	Visual Studio 2022

Note: The project is not compatible with Linux-based operating systems or macOS, as the included external libraries (GLEW, GLFW, SOIL2, Assimp) are compiled exclusively for Windows using Visual Studio.

Cloning the project from GitHub

The complete project, including the source code, 3D models, textures, shaders, and external libraries, is available in a public repository hosted on GitHub. The download procedure is described below.

1. Access to the repository

Using a web browser, go to the following URL:

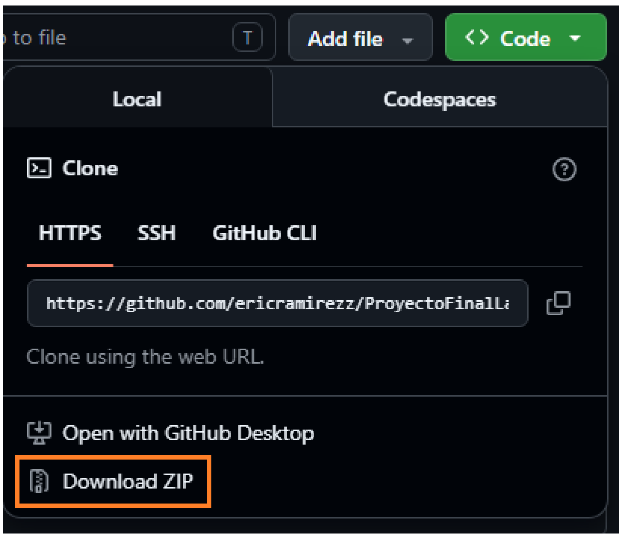
https://github.com/ericramirezz/ProyectoFinalLabComputacionGrafica.git

2. Selecting the main branch

Make sure you are on the main branch, which contains the final version of the project. Next, click the green “Code” button located at the top right of the file list.

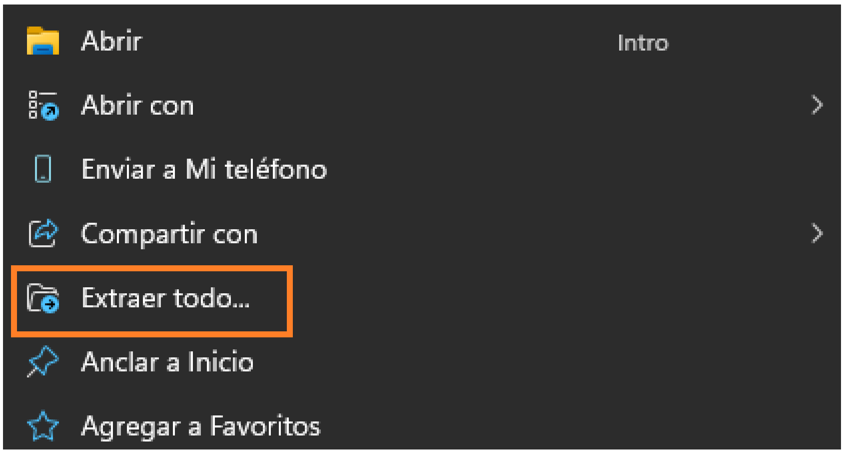
3. Download as a compressed file

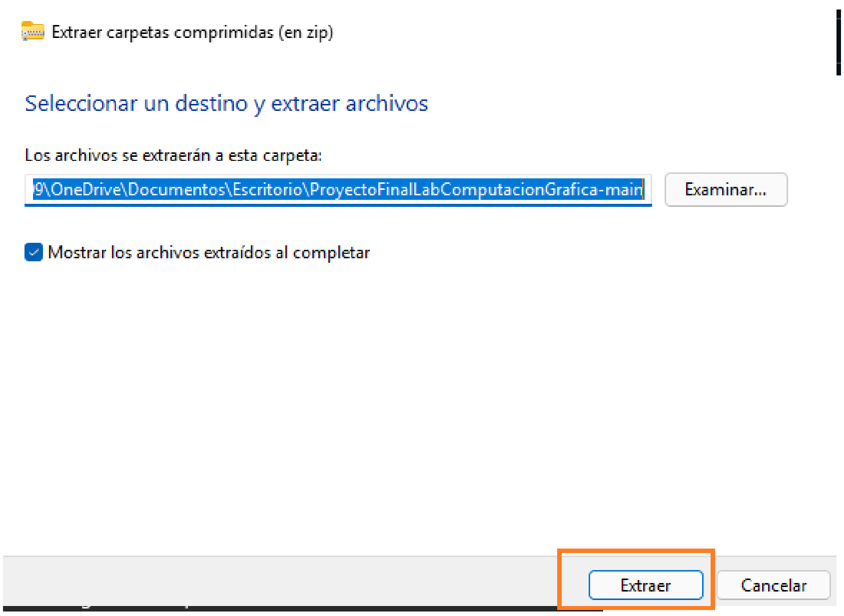
From the drop-down menu, select the “Download ZIP” option. A .zip file containing the entire contents of the repository will be downloaded.



4. Extracting the files

Locate the downloaded file in Windows Explorer. Right-click on it and select “Extract All.” Choose an accessible location and click “Extract.”





Once the extraction is complete, the following items should appear in the main folder:

- The Visual Studio solution file with the .sln extension.
- The project folder, which contains the source code files and resources (Models, Shader, Skybox, and others).
- The External Libraries folder, containing the precompiled external dependencies.

Running the project

Option A: Run directly from the .exe file (recommended)

This option is the fastest and does not require Visual Studio to be installed. It allows you to run the application directly, just like any other Windows program.

1. Location of the executable file

Inside the folder named “Executable” in the unzipped project, you will find the project executable. Click on the file named 320249538_PROYECTOFINAL_GPO_05.

This will start the system loading; in about 1 to 2 minutes, you will see the Engineering School bridge with its booths, lighting, and animated elements.

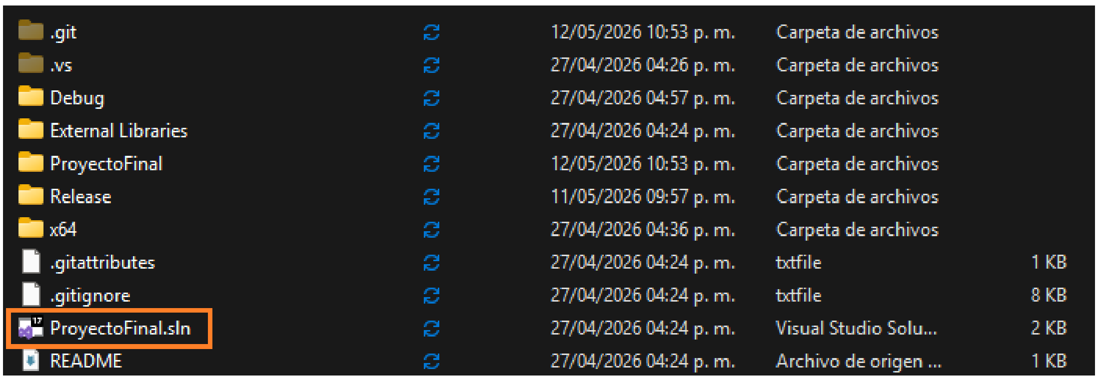
Option B: Compiling from Visual Studio

This option is intended for users who want to review the source code, make changes, or recompile the project. It requires Visual Studio 2022 to be installed with the C++ desktop development toolset.

Once the project has been extracted, open and run it in Visual Studio by following the steps below.

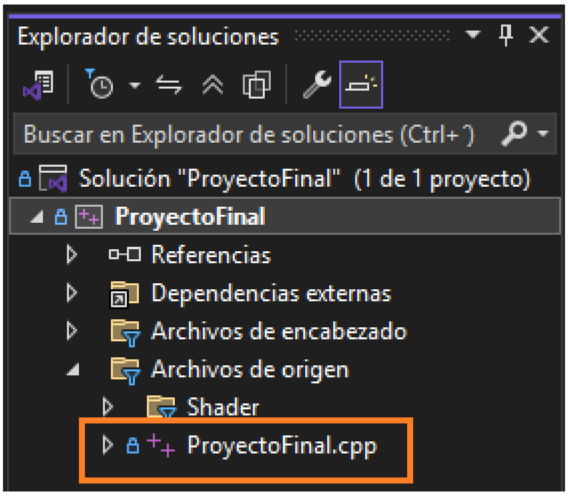
1. Solution Overview

Go to the project's main folder and double-click the ProyectoFinal.sln file. This will open Visual Studio and automatically load the entire solution.



2. Verification of the main file

In the Solution Explorer, located on the right-hand side of Visual Studio, expand the Source Files folder and verify that the ProyectoFinal.cpp file is present. This file contains the program's main logic: the rendering loop, model loading, lighting configuration, and animation handling.



3. Selecting the build settings

In the top bar of Visual Studio, check the two checkboxes as shown:

- Build configuration: Debug
- Build platform: x86

4. Build and run

Click the Windows Local Debugger button on the top bar, or use the F5 shortcut key. Visual Studio will compile the project, link the libraries, and open an application window displaying the rendered scene.

If the settings are correct, in about 1 to 2 minutes you will see the Engineering School bridge with its booths, lighting, and animated elements.

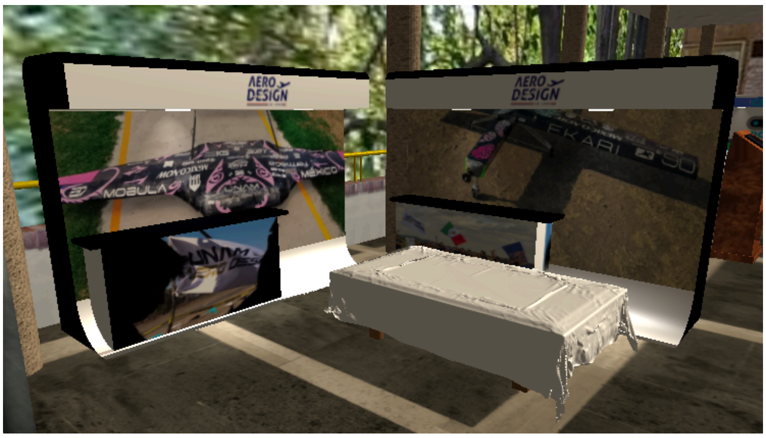
Description of the virtual scene

The scene shows the bridge connecting the buildings of the School of Engineering, a spacious, covered area used for fairs and exhibitions. The modeled elements are as follows:

- **Architectural structure of the bridge: deck, cylindrical piers, roof, and railings.**

- **Stands for sponsoring companies: Oracle, Amazon, and Procter & Gamble (P&G), each featuring a floating 3D logo and its own spotlights.**

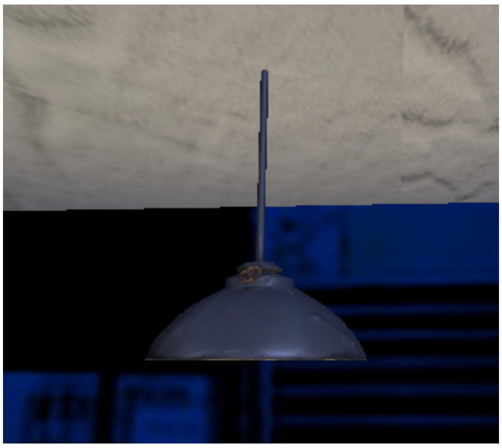
- **Student organization booths: CROFI, UNAM Aero Design, and SIAFI.**

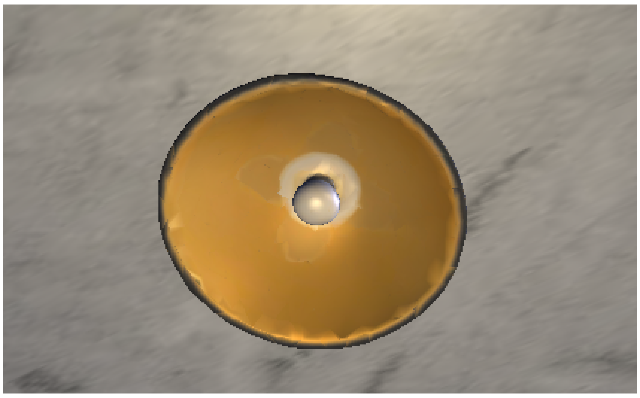


- Pumagua: a water station typical of UNAM.

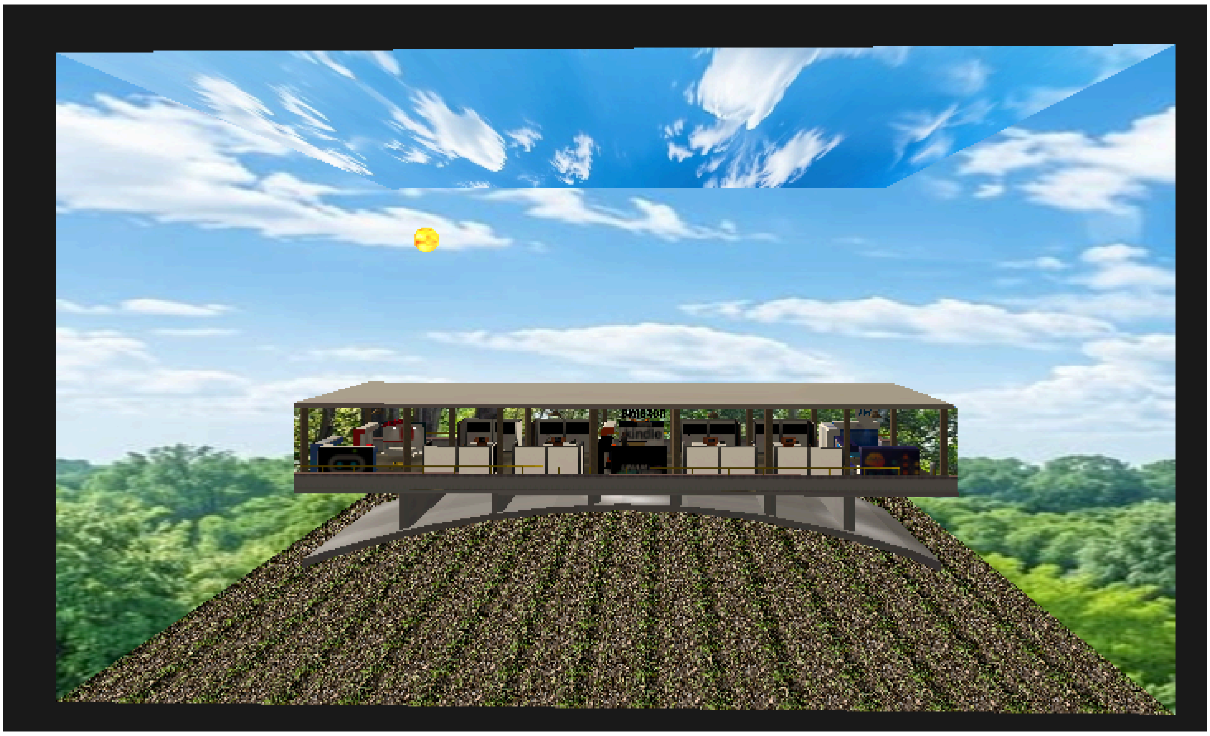
- A security camera mounted on one of the pillars, with automatic panning.

- Four ceiling lights spaced along the hallway, which are only turned on at night.





- Dual skybox system (indoor and outdoor), with separate versions for day and night modes.

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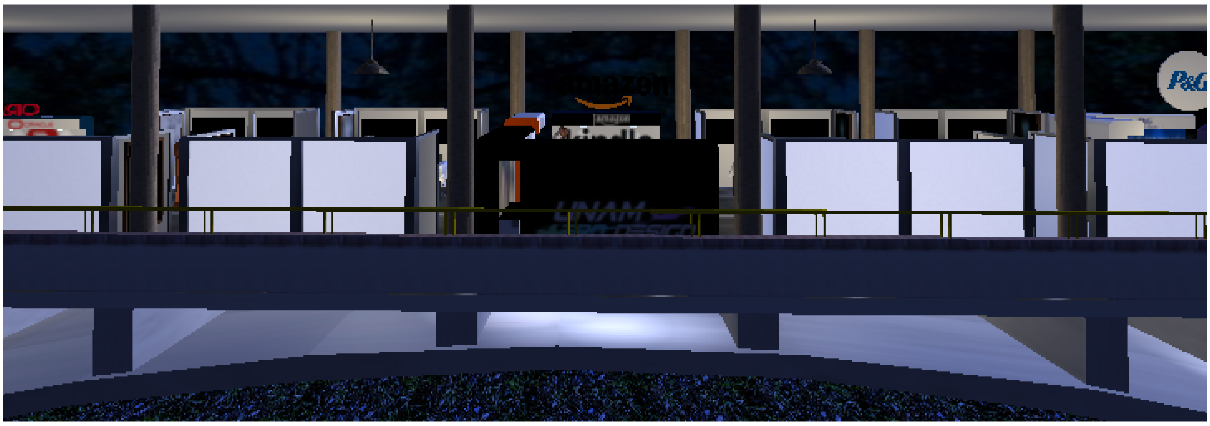

- **Emission models for the sun and the moon, which switch depending on the active lighting mode.**

Movement controls

The user interacts with the virtual environment using the keyboard and mouse. The camera operates in first-person view, allowing the user to directly control their viewpoint within the scene.

1. Camera movement

Use the following keys to navigate across the bridge. The arrow keys can also be used as an alternative.

- **W or ↑: Move forward.**
- **S or ↓: Move backward.**
- **A or ←: Move left.**
- **D or →: Move right.**

2. Camera orientation

The mouse controls the camera's orientation. Horizontal movement changes the rotation angle, while vertical movement changes the tilt. This allows you to view elements in the scene from any angle, including the ceiling, the sky, and the floor.

Day mode and night mode

One of the project's features is the dynamic switching between two lighting schemes: day and night. This change modifies the main directional light, the skybox textures, the ceiling spotlights, and the light sources visible in the sky.
To switch between the two modes, press the L key. The project starts in day mode by default.

1. Behavior during daytime

- The directional light source simulates the sun with warm, yellowish-white tones.
- The sun’s emission model is visible in the sky.

- The interior and exterior skyboxes display a clear, bright sky.
- The ceiling lights remain off to reproduce realistic daytime lighting conditions.
- The spotlight-type lights on the stands remain on at a reduced intensity.

2. Behavior during night mode

- The directional light takes on a cool bluish hue to simulate moonlight.
- The Moon's light source replaces the Sun in the sky.
- The interior and exterior skyboxes switch to their nighttime versions (starry sky).
- The four ceiling lights turn on, casting warm light onto the central aisle.
- The lights on the stands increase in intensity, highlighting the products on display.
- An additional spot light is activated to simulate the moon's reflection inside the bridge.

Project animations

The project consists of seven independent animations, each using a different implementation technique. The following table summarizes their characteristics.

Animation	Technique used	Activation
Visitor walking	Keyframes (10 frames)	Automatic
Host	Jerarchical + sinusoidal functions	E key
Robotic arm	Jerarchical keyframes (9 frames)	B key
Robot dog	Skeletal animation (FBX)	P key
Security camera	Continuous sinusoidal rotation	Automatic
Rotating logos	Rotation transformation	Keys 1, 2, 3
Flying brochures	Trigonometric + phase logic (flight / rest)	F key

1. Visitor Animation

An animated character, called a "visitor," continuously traverses the entire bridge. The animation is implemented using the keyframe technique and consists of ten keyframes that define the start and end points of each section of the path.

Behavior sequence:

1. Walk from the center of the bridge toward the far right.
2. Turn 90° to look at one of the booths and pause for a few seconds.
3. Turn another 90° and walk toward the opposite end of the bridge.
4. Repeat the pause to look at a booth on the left side.
5. Return to the center and repeat the cycle indefinitely.

The animation starts automatically when the program is launched and does not require user intervention.




2. Host's presentation

At one of the booths, there is an articulated model of a presenter that performs natural movements to welcome visitors. The animation is created using a hierarchy of three joints (forearm, hand, and head), whose rotations are generated by sinusoidal functions that produce a continuous motion.

The E key is used to activate or deactivate this animation. When deactivated, the joints return to their initial position.

3. Robotic arm animation

A robotic arm is positioned on a table at a booth. The animation is created using the keyframe technique and consists of nine keyframes that depict the complete cycle of picking up a part, lifting it, rotating it, placing it down, and returning to the rest position.

The kinematic hierarchy of the arm is as follows:

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fixed base → swivel base → shoulder → arm → gripper
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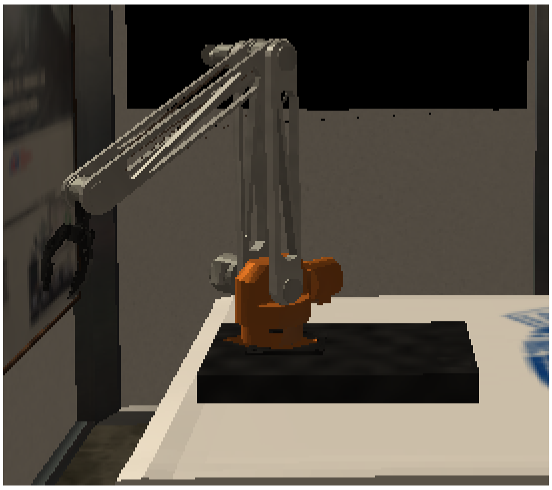
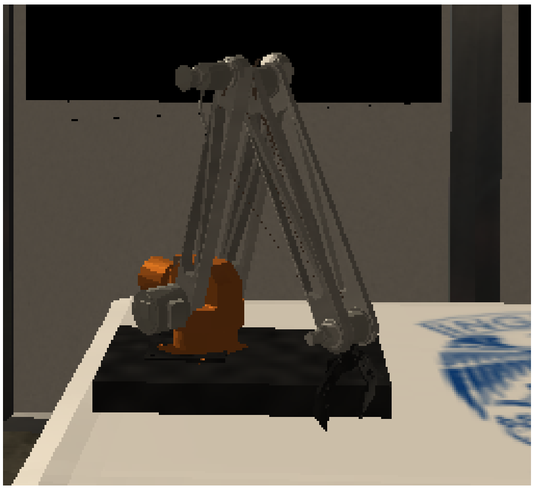
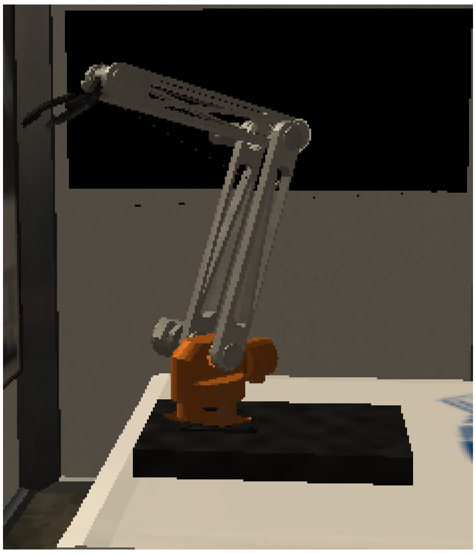
Each level inherits the transformation from the previous one, resulting in a coherent, fluid movement.

Use the B key to pause or resume the animation. When paused, the arm retains the exact position of the current frame; when resumed, it continues from that point.







4. Animation of the robot dog

The CROFI stand features a robotic dog animated using skeletal animation. The model is loaded from an FBX file containing both the mesh and the bone skeleton, and the animation is rendered in real time using a specific shader that transforms the vertices based on the bone matrices.

The P key is used to enable or disable the animation. The dog starts in a paused state; pressing the key causes it to start moving, and pressing it again stops it and restarts it from the first frame.

5. Security camera animation

Mounted on one of the pillars is a security camera consisting of a fixed base and a rotating head. The head performs a continuous panning motion with an amplitude of approximately 90° (from −45° to +45°), generated by a sinusoidal function with a period of approximately twenty seconds.

This animation runs automatically when the program starts and remains active indefinitely. It requires no user intervention.

6. Logo rotation animation

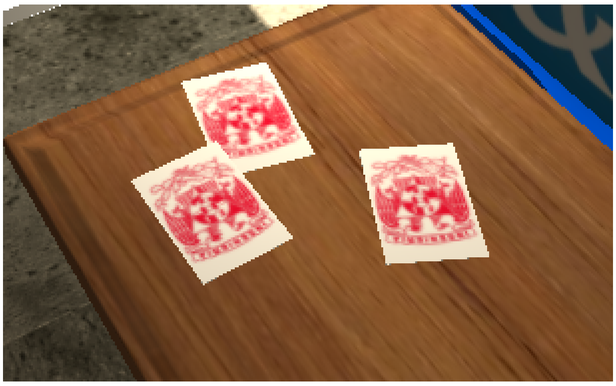
A three-dimensional logo hovers above each of the company's booths and can rotate independently around its vertical axis. Each rotation is controlled separately using a number key:

- Key 1: Rotate the Oracle logo.
- Key 2: Rotate the Amazon logo.
- Key 3: Rotate the P&G logo.

When you disable any of the rotations, the corresponding logo reverts to its original orientation.

7. Animation of flying brochures

A static stack of promotional brochures rests on the table at the P&G booth. Automatically, a couple of brochures periodically fly from the stack onto the bridge floor, following a curved trajectory with multiple rotations during flight. Upon landing, each brochure settles in a natural orientation. The animation plays continuously when the program starts. The F key allows you to restart the cycle if you want to watch the start of the flight again.



Keyboard Shortcuts

The following table lists all the controls in the program, along with their associated actions and their initial state when the application is launched.

Key	Action	Initial state
W / ↑	Move the camera forward	—
S / ↓	Move the camera backward	—
A / ←	Move the camera to the left	—
D / →	Move the camera to the right	—
Mouse	Orient the camera	—
L	Toggle between Day mode and Night mode	Day
E	Enable/disable the presenter animation	Active
B	Pause/resume the robotic arm	Active
P	Enable/disable the robot dog animation	Inactive
1	Rotate the Oracle logo	Inactive
2	Rotate the Amazon logo	Inactive
3	Rotate the P&G logo	Inactive
F	Restart the flying brochure cycle	Active
ESC	Close the app	—

Exiting the Program

To finish running the project and close the application window, press the ESC key. Alternatively, you can close the window using the standard Windows close button. Visual Studio will remain open so you can compile and run the project again whenever you like.